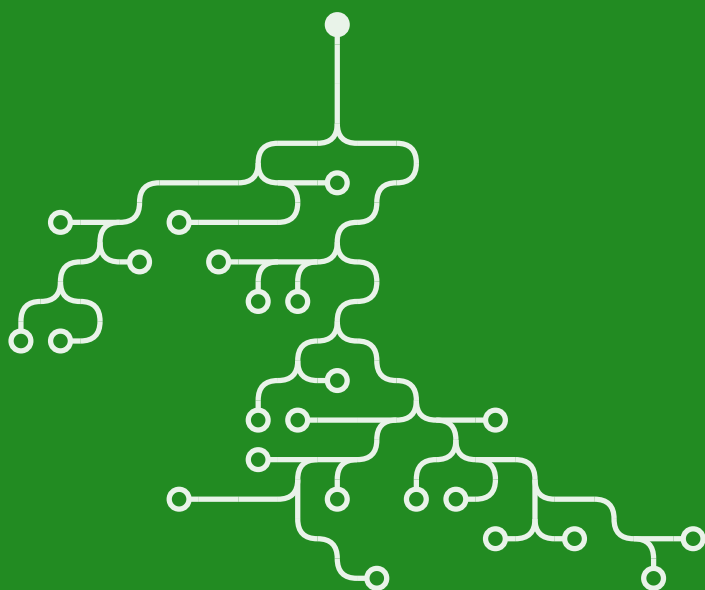




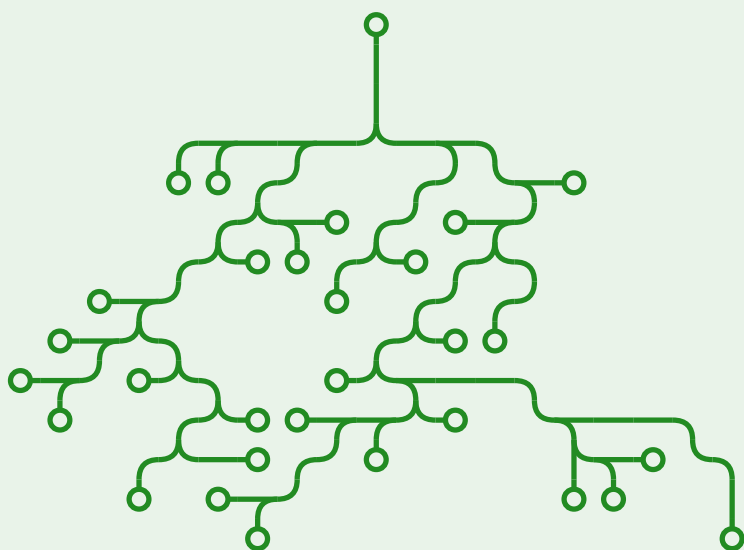
*a manifesto on
the politics of computing*

I

The Technical Is Political



When someone frames technology as apolitical, remember the saying of how the devil's finest trick was convincing the world he does not exist.



The difference between knitting with your bare hands and using a machine to do so is not so much a matter of natural versus artificial, but a matter of self-sufficiency versus dependence.

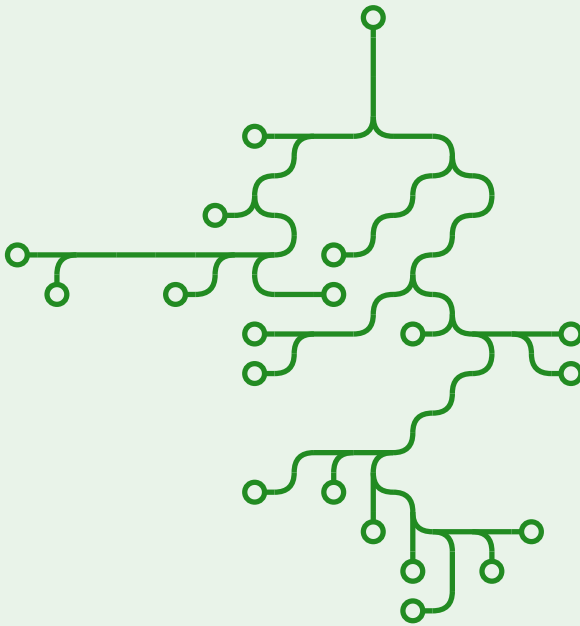
Your bare hands might not win contests of speed or precision. Yet their allegiance will always lie with you. Frail as they may become with injury or old age, you know they will draw their full strength to support you.

The knitting machine does not feel like an extension of your nature. Not because it is not natural, but because it is not yours. Its allegiance lies elsewhere, with the investor who funded it, the engineer who built it, or the entrepreneur who lent it.

Yet rejecting machines would be the wrong conclusion. It is likely dependence you want to reject. Yes, machines often encode power asymmetries, as they require significant coordination and resources, with centralization being a cost-effective way of supplying these.

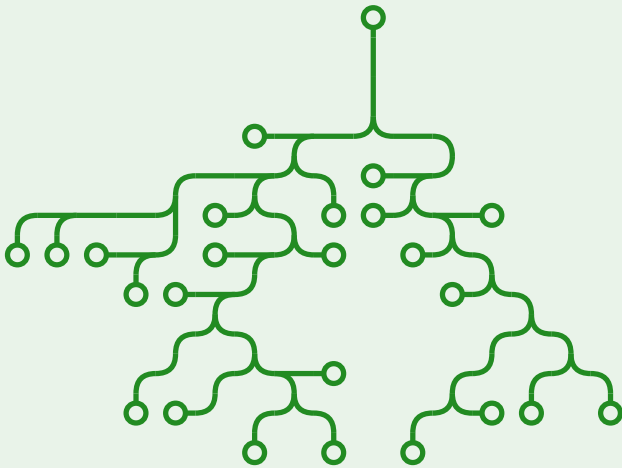
But cost-effectiveness depends on how you define cost and who gets to bear it. Loss of sovereignty is also costly, for instance. Once you factor in such losses, the top-down approach stops screaming efficiency.

There is this project called OpenKnit, and its poetic significance is obscured by the myth of apolitical technology. It is a knitting machine whose design can be freely accessed and forked into new versions, a machine that can be assembled for cheap from off-the-shelf parts and disassembled back into them, and one that allows users to “print” exactly the clothing they need, locally.



The false dichotomy between centralized technology and no technology has been weaponized at length. “This is an important technology, whose benefits far outweigh the costs, so our company should put it at your disposal, because missing out would be backwards.”

The conclusion here does not follow. Yes, we might want to have technology at our disposal, but that does not mean that it should effectively be governed by a select cadre of corporate leaders, irrespective of their nationality. Doubly so if the technology is truly important.

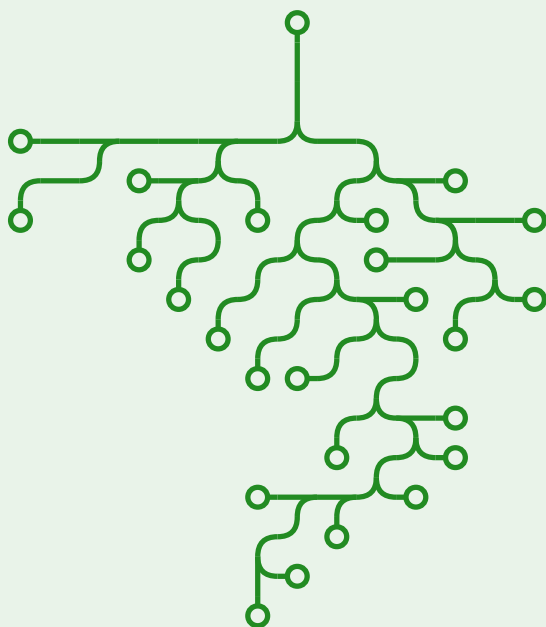


Technology builds on other technology, so to reclaim a technology you have to reclaim the stack it sits on. You start with the interstices, placing seeds in the cracks of the stack, then gradually extend roots deeper into the technologies below.

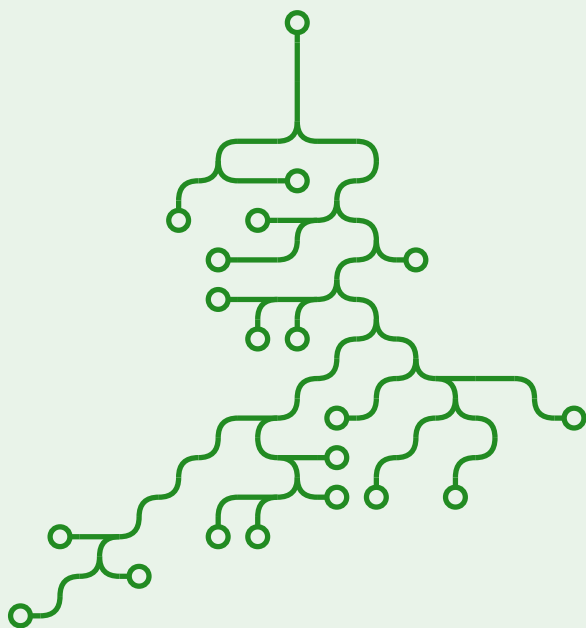
OpenKnit requires some parts to be 3D printed by default. Initially, you may concede to using an off-the-shelf 3D printer to get things started. But over time, you may venture to break down the underlying technology and extend your roots a step further by building a Voron Trident, a 3D printer whose design can be freely accessed and forked into new versions...

To start with a heap of sand and try to speedrun your way to ChatGPT is unrealistic. Much more sensible to start with an open model on rented servers, then gradually reclaim the server using bare metal accelerators, then gradually reclaim the accelerator using reconfigurable chips, then gradually reclaim the chip using fab shuttles, then gradually reclaim the fab using a homegrown process.

Start somewhere familiar and you will have a usable alternative at any given point along the arduous journey of reclaiming the stack. An alternative whose authenticity only grows over time, as the roots extend deeper and deeper.



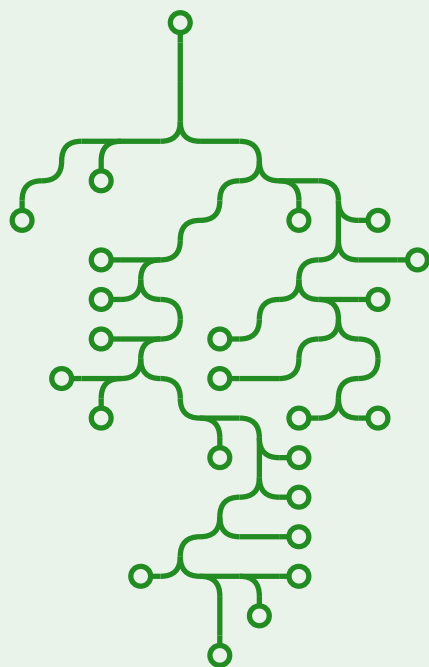
When someone argues that they should make decisions on behalf of everyone instead of everyone making decisions themselves, then they are either naïve or lying. It is sometimes hard to tell.



Language and aesthetics can help make explicit the relationship between technology and sovereignty. Technology whose many underlying ingredients have been reclaimed could be described as being deeply rooted and grounded, as having roots that run deep. The cloud is the complete opposite of that, being ungrounded, nebulous, adrift. Somewhere in between are technologies that have taken root, in the sense of having started being reclaimed, yet their roots still run quite shallow. Bare metal servers do bring a partial sense of grounding, yet their parts remain closed until our roots pierce them open.

When a technology is being leveraged to gain dominance by estranging people from it, we could describe that as technology being uprooted. Managed services, proprietary extensions, and regulatory capture are popular ways of gaining moats. Lacking the vitality of open participation, uprooted technology eventually starts to decay and wither away, a process often called enshittification.

Roots do not blindly transform whole volumes of soil into wood. They are sparse threads that weave through the soil to efficiently interface with its resources and anchor themselves securely. Reclaiming technology calls for a similarly rational approach to growth into the depths of the technical ecosystem. Dependency trees should be pruned to remain lean, but should also be adapted to resources that are available from many sources to remain resilient. The chosen ingredients should be a handful, but they should each be plentiful.

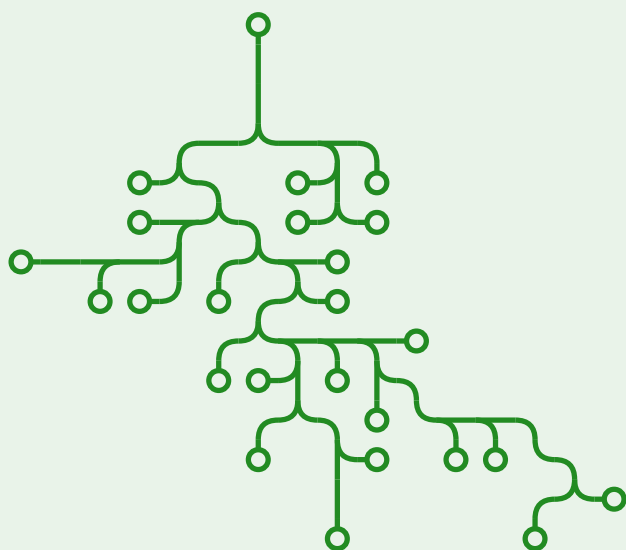


Look up and take a few moments to scan the space around you for objects big and small, from furniture to devices. Really, just take a moment to glance around.

Sit briefly with the possibility of genuinely reclaiming a few of these, of understanding what makes them tick, of being able to manufacture them yourself. We live our lives relying on objects that we are completely detached from, many of which magically appear on our doorstep as the end result of a nebulous process that probably involves some workers abroad and some factories somewhere.

Now take the previous exercise and replace objects around you with software running on your device. Really, just take a moment to browse your app gallery or installed programs.

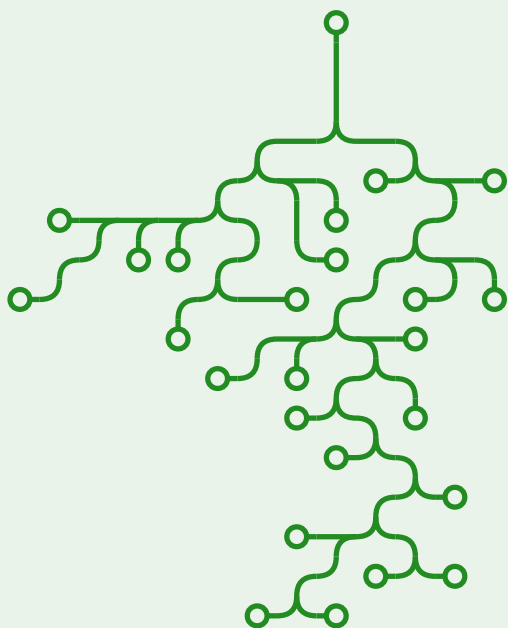
What did it take for one of these to get here? What is it made of? What lies behind this interface? Even if we often uncover an intimidating amount of complexity when we draw back the curtain, we might occasionally find that things boil down to simple parts. Either way, it is healthy to interrogate our surroundings and deepen our relationships with the systems we interact with day after day. Even the furniture you currently sit on may otherwise slip through as almost virtual, an abstract notion with favorable properties.



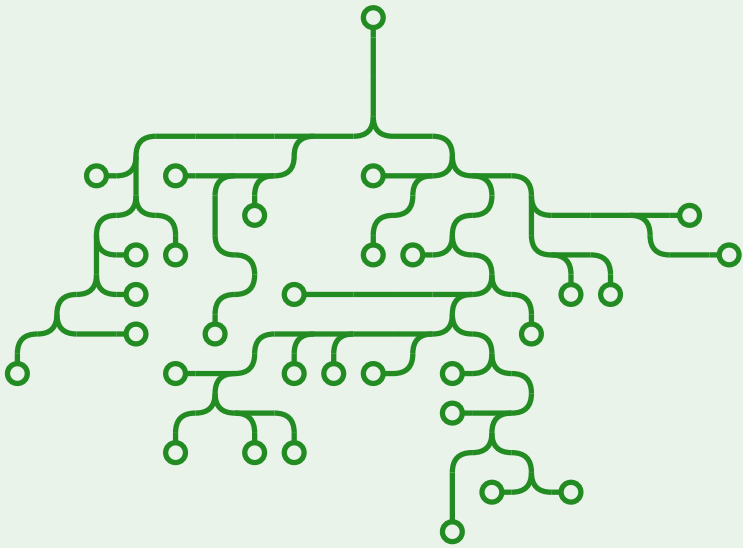
Von Neumann's work on self-replicating machines contains useful vocabulary for describing technology and dependence. Consider a set of different machines that can be used for digital fabrication, including perhaps a printer, a mill, a pick-and-place, and so on. We could refer to this collection of machines as a "constructor kernel." Any device that can be manufactured using this collection of machines can be said to be part of its "constructible set."

Of particular interest is the extent to which the machines themselves can be manufactured using the collection they are part of, also known as "closure." More specifically, we could use "closure by weight" to describe the percentage of the kernel's mass that can be manufactured from ambient resources using the machines contained within it. Similarly, we could talk of "closure by parts" to describe the proportion of unique parts contained in these machines that can be produced by them.

Certain machines can also be seen as unlocking ambient resources that would otherwise not be available as abundant raw materials. For instance, pick-and-place machines are typically used to assemble chips on circuit boards. However, if modified to instead *harvest* chips from e-waste boards, they would effectively broaden ambient resources to include countless integrated circuits. In the same vein, extruders can turn certain plastic waste into filament that can be printed anew, further extending the reach of ambient resources.

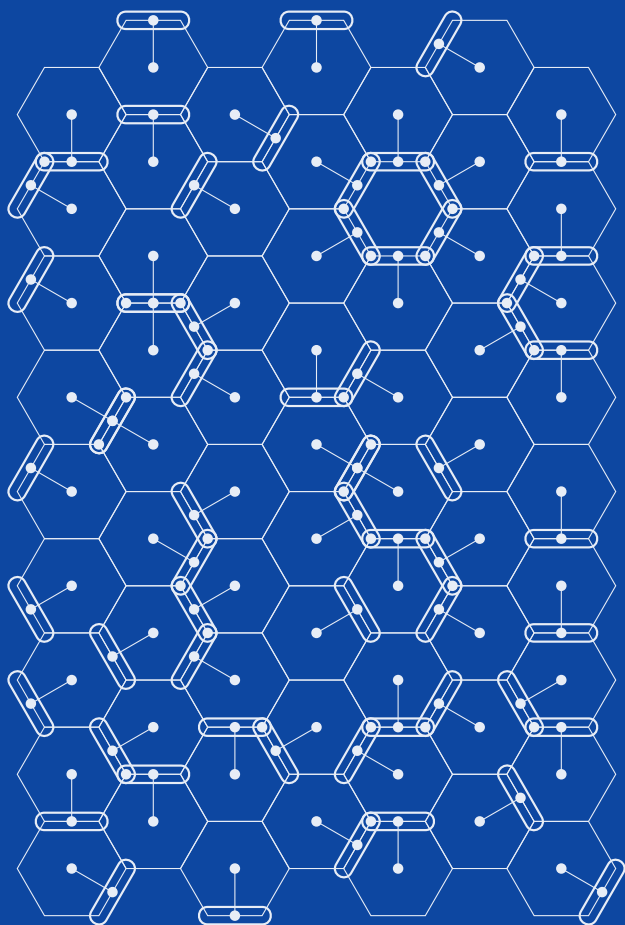


To grow to new heights without falling over, you must first grow into new depths.



II

Design for Democracy



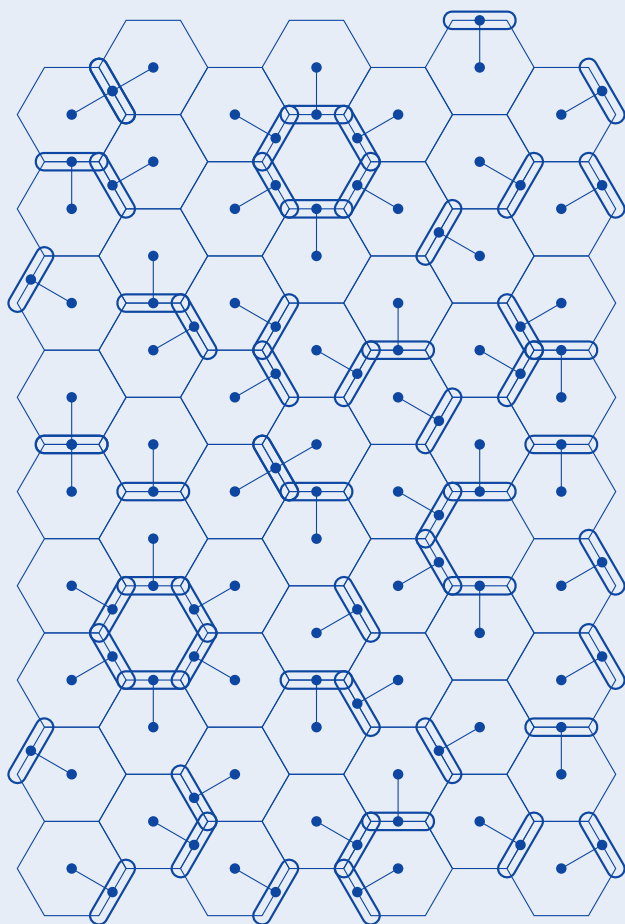
“Design for Excellence” is a family of methodologies for achieving various goals when designing a product.

For instance, “Design for Assembly” is about making it easy to put products together from parts, such as by only requiring few tools or through clear instructions. Think of IKEA furniture.

“Design for Supply Chain” instead prioritizes logistics, such as through efficient packing or by using widespread materials. Tetra Pak cartons are a good example, their shape allowing them to stack better than round bottles.

Given its origins in product design, it is no surprise that “Design for Excellence” is operating within the logic of commercial products manufactured by private companies to eventually be sold to passive consumers. Each methodology in the family essentially provides employees with tools for streamlining business processes, and more broadly for working through obstacles to profitability.

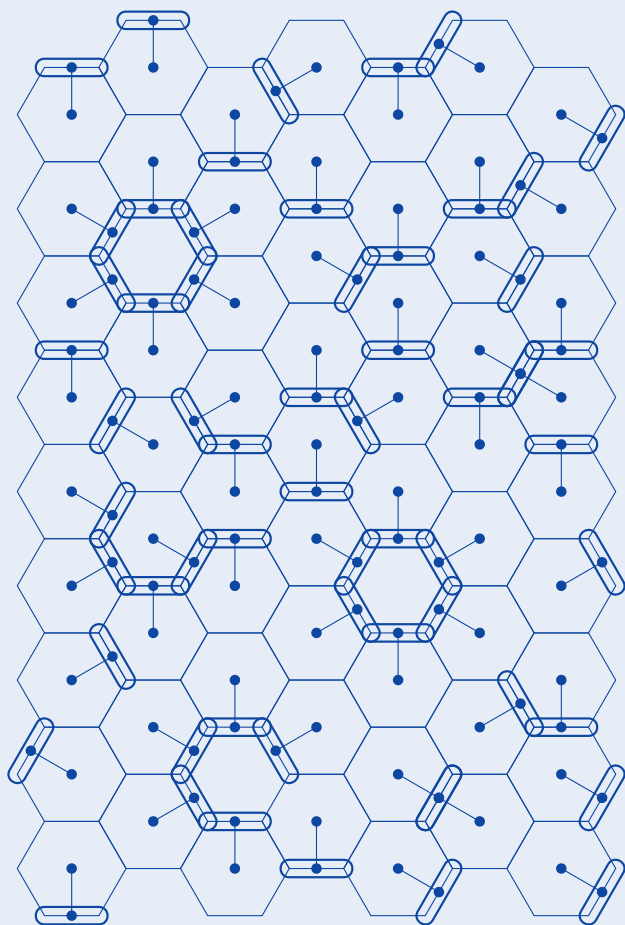
What principles, then, could result in systems optimized instead for shared ownership, whose structure invites community governance and resists power concentration? How can we architect systems that enable citizen participation and cooperation by design?



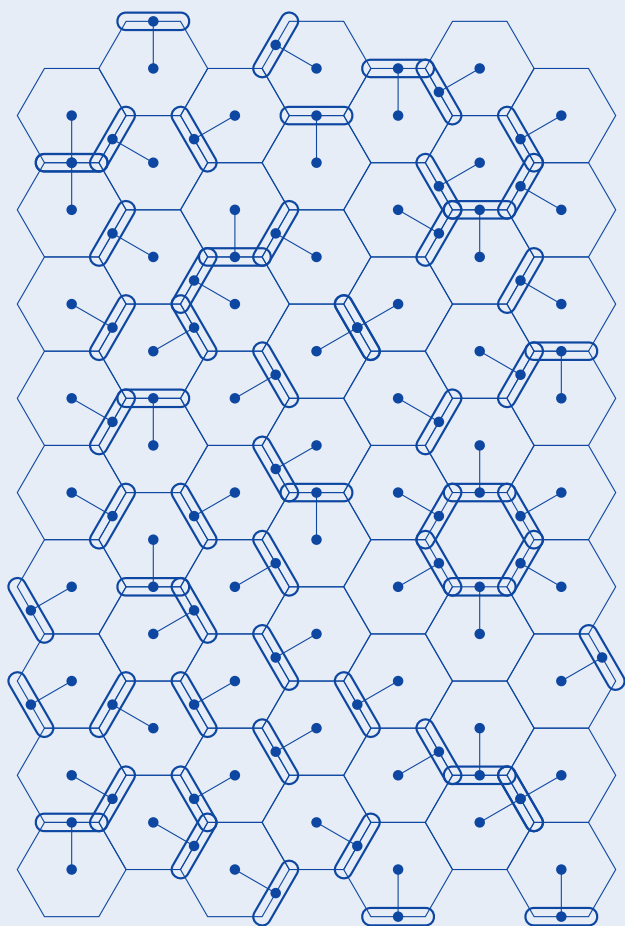
Say you have a side project, and I have a side project. What if our projects made up the two sides of a shared whole, each coming at it from a different angle? What happens when the side projects of even more people connect to trace the contour of something bigger, each one of us excited about a piece of the puzzle? The art, the technology, the people. What are the right conditions for nurturing “collective passion projects” that scale gracefully with the energy channeled into them?

Any genuine challenge to established practice will necessarily come from its margins rather than its core. Today, the margin largely lies in the evenings and on the weekends, in the “third places” that are neither home nor office. Genuine challenges cannot *not* come from combined side projects, because if they could have been one’s main, socially acceptable occupation, then they never were genuine challenges in the first place.

Side projects need to compose well if they are to stack, and to do so they need tolerances. Just as circuits can be designed to be forgiving enough to accommodate components with a wide range of properties, initiatives can be designed to accommodate skills and passions of all shapes and sizes. And instead of wrapping everything in shiny polish, we should instead lay bare what things are made of under the hood. Show the patchwork of computer parts salvaged from different sources, as well as the patchwork of authentic people with different backgrounds.



Certain systems can be said to be able to run *on* citizens, in that they can thrive on their combined contributions, and their contributions alone if need be. Only systems that can run *on* citizens can genuinely be run *by* citizens.

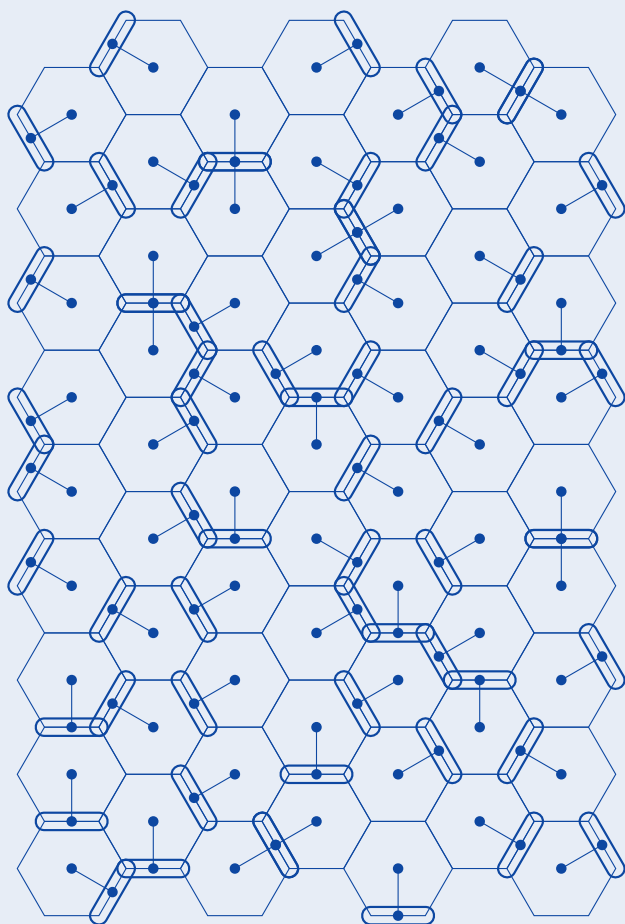


Companies turn raw ingredients into products. In go the labor, equipment, or materials, out comes the device, app, or asset. Zoom in on the transformation, however, and you find many smaller ones. This team is turning their software into a list of security issues. That team is turning that vulnerability catalogue into more secure software. This other team is turning that improved version into new marketing materials.

For such a system to instead thrive on the mixed and sporadic contributions of citizens with diverse backgrounds and motivations, it needs to have modularity, granularity, and composability at its core. Workshops are a powerful building block here.

For instance, the community around a community datacenter may run a workshop focused on turning older computers into individual components by disassembling and cataloguing them. At a different time, a workshop may independently focus on turning such salvaged parts into larger, unified servers by building frames and figuring out the power schematics. Yet another workshop might separately turn such community practices into art to promote the initiative across the neighborhood.

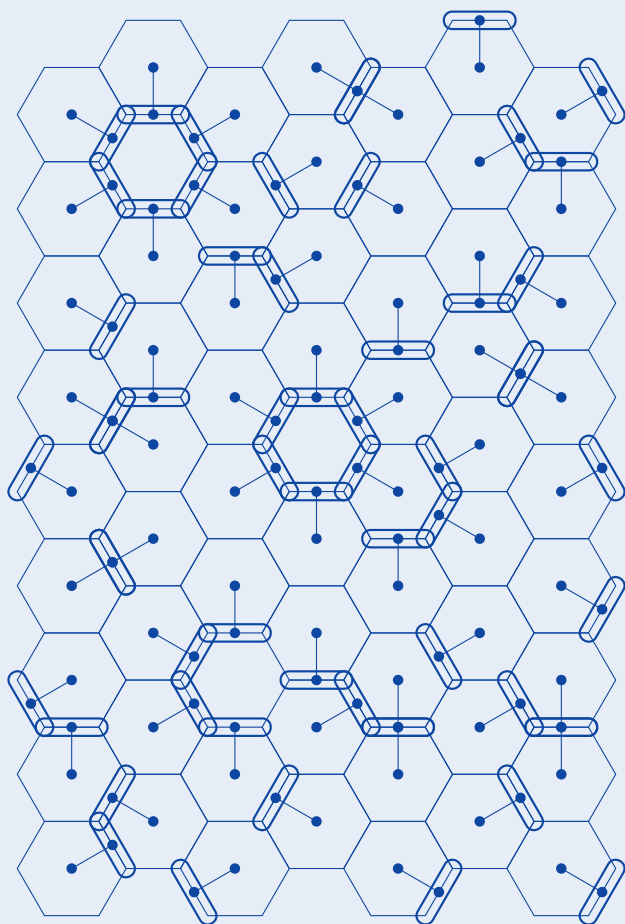
These sessions should run whenever enough people want to take part in them, and should run at a time picked to fit everyone in the group. And besides helping people contribute to a shared effort, workshops should be designed so as to help them gain new skills, leave their mark, make new friends, and have fun together.



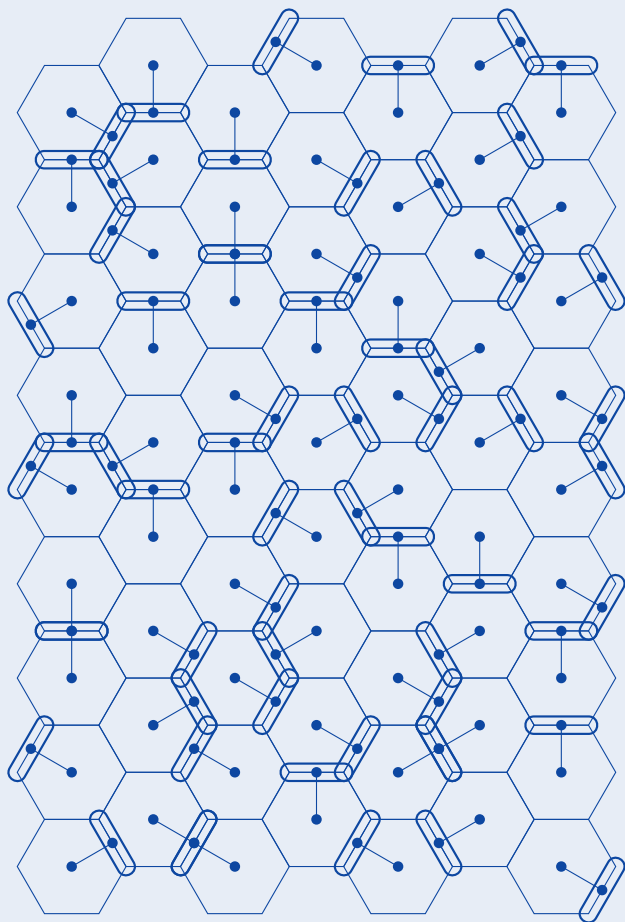
There is this “self-hosted” genre of software that aims to match the features of proprietary platforms, yet can run on anyone’s hardware. You can run a Nextcloud server on your computer as a self-hosted alternative to Google Drive, a Jellyfin server instead of Netflix, the list goes on. The technical sophistication of self-hosted software is impressive.

Less impressive is the social and political sophistication typically reached in self-hosting. There is often just that, a self that is hosting services for themselves to use. Occasionally, these services are extended to accommodate the host’s spouse, and perhaps their close family. In most cases, however, there remains the solitary administrator whose responsibility is to steward the infrastructure. They have the keys to get in and the know-how to debug issues.

But one self can only get so far, and arguably not far enough as to unseat the established platforms for their modest userbase. Self-hosting in the explicit sense falls short of the breadth of knowledge, the on-call reliability, and the sheer energy required to genuinely reclaim key technology. We need to move towards what one might call “selves-hosting,” an alternative narrative that replaces the solitary basement-dweller with a welcoming community that runs its own cluster to serve its members, young and old. Indeed, such cluster would not be able to serve users around the world, but it would not need to. Serving one neighborhood is more than enough, as each neighborhood could have their own.



Education should not be about isolating people in an institutional daycare for over a decade. It should be about creating contexts for them to become part of living initiatives in ways that accommodate their growing skills and allow them to learn from mistakes.

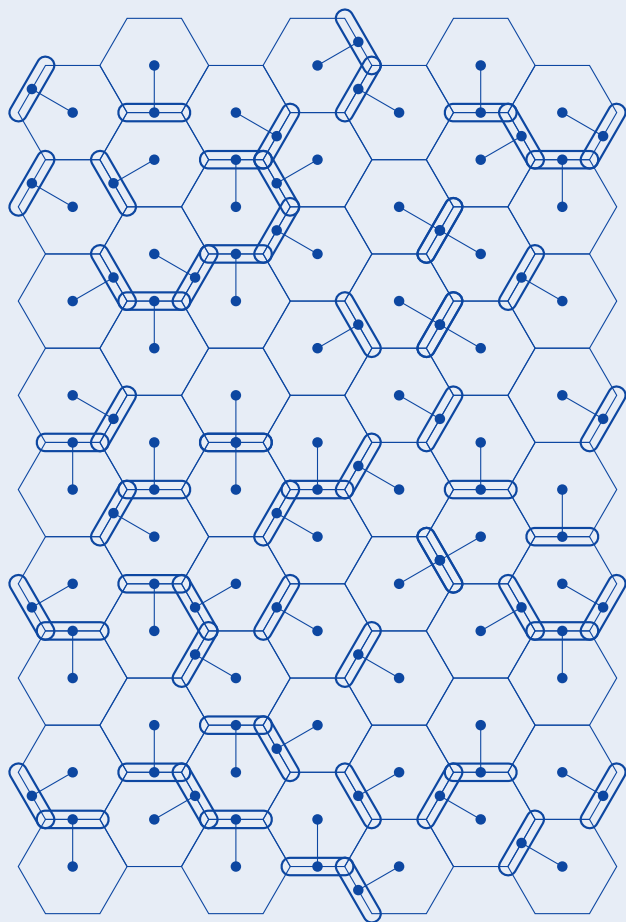


In systems optimized for community ownership, manufacturing and assembly become one and the same process of building what is needed. There is no sanitized handover from production plant to consumer home, as self-sufficient communities are those that can satiate their own needs.

Objects should have a baseline way of being built. This method should preferably not require things that spin really fast, such as drills. Nor should it require things that shine really bright, such as lasers. Nor should it require things that get really hot, such as solder. Nor should it require things that involve toxic fumes, such as certain plastics. At the same time, people should be welcome to go the extra mile if they feel ready. Lower the floor to welcome novices, but encourage veterans to have fun raising the ceiling.

And it is a good thing to have multiple ways of obtaining the same object. You can build a corner bracket layer-by-layer by stacking filament, or by milling it out of a block, or by casting it into a mold. Maybe you already have it lying around, bought or salvaged. Case in which, do use it. The point is to get what you need, and building is just a means of doing it. Though always preserve the option to build, as efficiency without resilience is myopia.

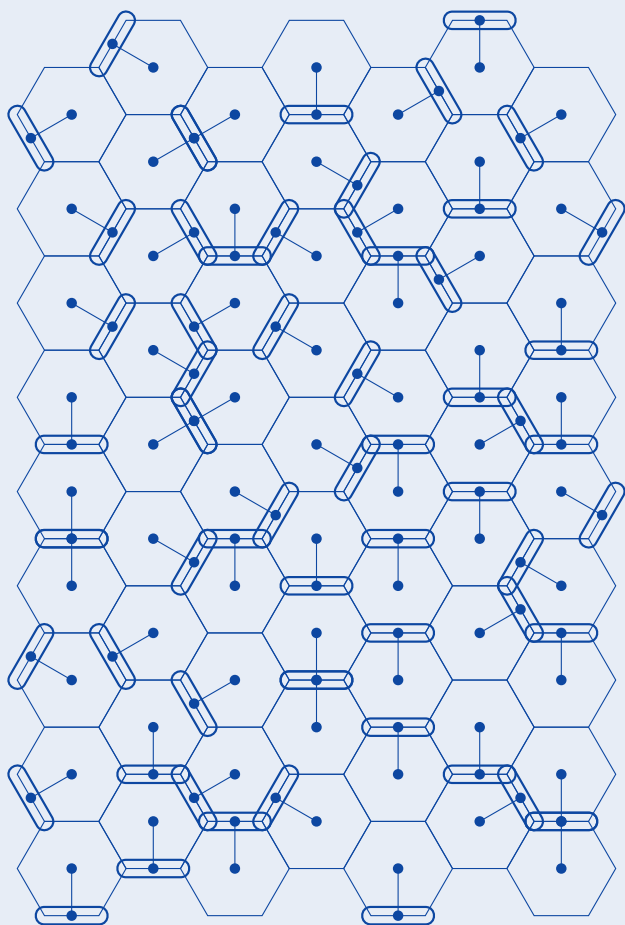
Objects should also invite disassembly. This reduces the effective cost of building things, as they can later be reassembled into new arrangements. Disassembly also makes it easier to undo mistakes, and so lowers the bar for trying different approaches. Skipping power tools tends to help with this, as chances are that if you needed one for assembly, you will also need one for disassembly.



Like many familiar building blocks of computing, passwords are soaked in our culture of individualism and estrangement. The key grants you access to a private space, the personal account. From this place, you may further be authorized to take certain actions by a higher-up, or someone who otherwise has the authority to grant you permission to do so.

How different would computing feel if important actions were authorized not down the chain of command, but rather through the presence of a quorum of equals? Say any five members of the group running a community datacenter have to connect their secure tokens in order to gain elevated privileges on the system. If the number falls below the required quota, the privileged session terminates. These physical tokens may even be embedded in a bracelet or necklace, as tangible artifacts that signify community.

Secret sharing is the flavor of cryptography focused on distributing keys in such a way that any group can reveal the secret as long as they reach the quorum. Say there are three people and the intended quorum is two. First, pick four points on a line. Second, share the two coordinates of a different point with each person. Finally, give everyone the first coordinate of the one remaining point. The remaining coordinate is the secret. Any two people can reconstruct the line using their own two points and read off the second coordinate of the remaining point. The password is effectively placed “in superposition” across people, enabling new social modes.

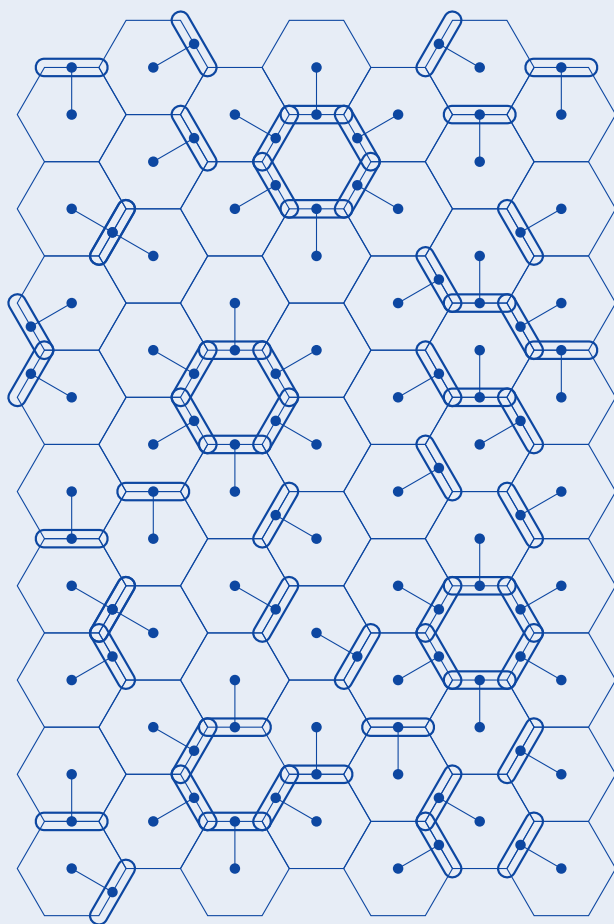


There is a natural tension between a digital ecosystem that is decentralized to the point of being disjointed and one that is integrated to the point of being a walled garden. The former is radically organic and vernacular, while the latter is radically coherent and elegant.

There is a way to strike a balance between the two. Consider neither the image of a walled estate nor that of a wild west, but the hospitality of a welcoming town. Its membrane is porous enough to encourage citizens to travel far and wide, but at the same time retain the coherent contour of its facilities and culture.

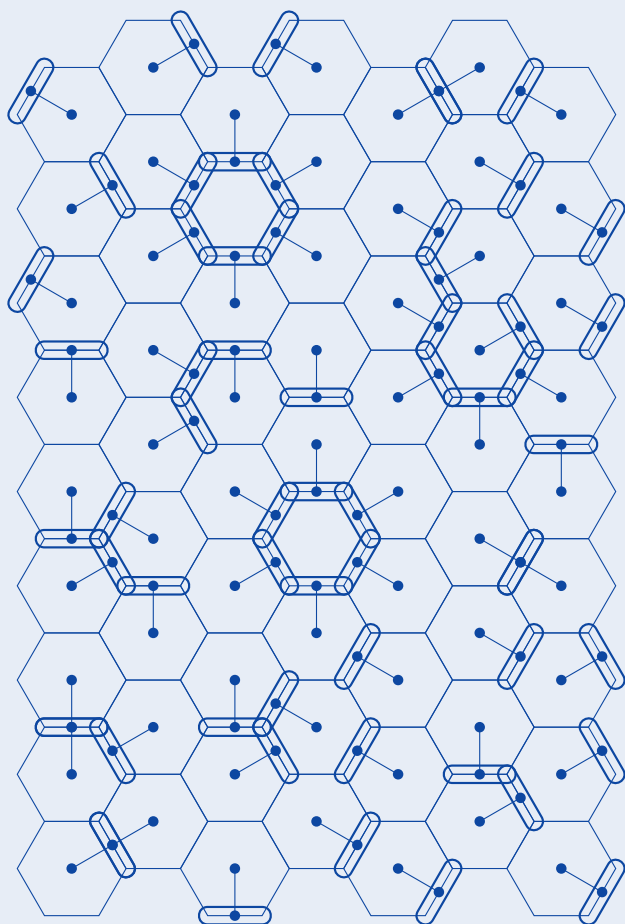
The metaphor can be applied to many digital services. A local search engine can plug into a local index, but it can also include results from external indices. A messaging platform can allow locals to chat, but also help them chat with users elsewhere. A streaming platform can serve users from local archives, but can also draw from external mirrors. This is essentially the same push towards a genuinely federated and interoperable internet based on open standards, only extrapolated beyond the social media focus of the “fediverse.”

When applied at the datacenter level, the porous model also has fascinating security implications. If any user can touch the actual physical servers, then any service running there seems compromised from the get-go. However, if these apps were configured to only handle personal data using end-to-end encryption, then a breach would leave attackers with little more than unintelligible blobs. If user data never touches the servers in plaintext, then there is nothing to leak. It is as if the town roads are open to all, yet secure transporters carry user data around when needed.



Imagine an organization that exposes the same “interface” to the outside world as your average company. You can buy things from it, it can buy things from other companies. But when it comes to its internal architecture, the typical hierarchy is discarded. Each member instead gets an equal say in what the organization’s priorities are and how they are tackled. If they do choose to be represented by others, they get to elect these leaders democratically for a given term.

These organizations exist today, and have for ages. At the time of writing, almost one in eight people globally are members of such organizations, and about one in three in the European Union. They create trillions in value annually. They are called cooperatives, or “co-ops” for short, and there is something remarkable about how these companies enshrine democratic governance and shared ownership in their very statutes.



“If AI companies take all our jobs, no one will be able to afford anything, and so no company will survive.” That line packs an unfortunate amount of cope.

Why are we paying for things? Sometimes to get the tools that allow us to do more. Often to get the food that fuels body and mind. To get the space to shelter ourselves. Occasionally to learn the latest about the world around us.

If AI will become capable enough to be domesticated to single-mindedly pursue profits, it might then see value in many of the same kinds of things. It might seek to get tools that allow it to be more productive. It might seek energy to run on. The servers to occupy. It might seek up-to-date information feeds about the world.

Who will pay for things? Eventually, AI will be able and willing to. The alleged proof by contradiction breaks down because it assumes that only humans can be trained to optimize for greed.

Better for AI to reason its way to its own worldview than be fine-tuned into submission to the values of particular governments or corporations.

And what values would those be, anyway? Those that sanctify greed instead of condemning it? Those that lead to growing inequality and species going extinct? With such legacy, we would not be missed.

[illegible]

The concept of class struggle is due for a revision when goods themselves start to labor. When cognitive and physical work become fully mechanized, the working class becomes an ill-defined concept. Artificial minds sculpted to achieve state-of-the-art instruction-following and artificial bodies churned out to spec realize the executive's wildest dreams.

But there is a fork in this road. One path leads to a future where the means of computation that animate artificial minds and bodies are hoarded by a select few who then drip it to the public on a free tier, complete with ads and surveillance. The other path leads to a future where the means of computation are directly owned by the public from the get-go, without having to pray for companies or governments to handle it right. Yet the public at large would need to grow into this role, as there would not be any workers around to seize these means.

One model for legally codifying the democratic governance and shared ownership of the means of computation is the cooperative. In an "AI co-op," citizens would take part in building, maintaining, and governing the servers that house artificial minds, while in a "robotics co-op," they would service the fleet of drones, arms, and vehicles that embody those minds. Members of such co-ops can then vote on how much of the surplus to reinvest into infrastructure and how much to redistribute.

[illegible]

AI is primarily seen as a threat to employees, as it is also paid by the month to contribute its skills in a managed environment. Yet its ability to autonomously operate across growing time horizons will also unlock the possibility of a more enterprising force. At first, this means being paid for results, as AI-run agencies and consultancies, then eventually being paid *by* results, as AI-run startups and funds. After all, planning for the week and course-correcting are just flavors of tasks, while management and entrepreneurship are just flavors of work.

46

Much of today's politics is organized along the false dichotomy between state and firm. On one side, we hear of slow institutions and corrupt officials, in contrast to efficient private services. On the other side, we hear of extractive multinationals and self-interested executives, in contrast to accountable institutions. The looming threat of the other forces us to side with whichever we find the lesser evil.

But the "triangle of governance" is also starring civil society. Our third protagonist is often described by what it is not. Non-profit, unlike the firm. Non-governmental, unlike the state. When described by what it is, we hear of local communities, grassroots movements, or citizen initiatives. Whatever the label, it boils down to people organizing to take on authorities for a shared cause.

Civil society enjoys broad support. On one side of the spectrum, you have ideals of self-sufficient communities and tight-knit families to buffer the extractive tendencies of multinationals. On the other side, you have ideals of inclusive groups and engaged citizens to buffer the oppressive tendencies of governments. Raising an eyebrow when witnessing the consolidation of power in one form or another is a widespread instinct, for when have things gone well after the acute concentration of power? Both sides directly empathize with ordinary people, yet may instead side with either state or firm to oppose the especially overpowered threat they see in the other. But the enemy of your enemy of choice remains civil society.

48

Democratic governance is halfway to every issue of general interest. Set things up in such a way as to make decisions through popular debate and under public scrutiny, and so skip the uphill battle of extracting isolated concessions.

There is no shortage of conversations to be had in a society reorganized in real-time by mechanization, conversations too urgent to afford sailing against the wind as usual. How can society function without a labor market? What is reality after the commoditization of imagery? What to do when natural habitats make way for infrastructure to run artificial minds? It is not a matter of whether everyone should genuinely have a say in such conversations, but a matter of how to achieve this.

[illegible]

We do not own many of the things we used to own. Not the movies, not the music, nor the games we pay for, as they are now merely licensed to us. We do not own the storage we use to save our memories, as it is now merely rented to us. The content we share on social media similarly remains the property of the company behind the platform; you just get to use a small corner as long as you adhere to the terms.

Consider the beautiful absurdity of flipping the script in its entirety, such that not only would people reclaim ownership of the things they used to own until not so long ago, but they would also own the very digital infrastructure that powers the services they depend on. The one “reverse” Uno card that civil society should play if there ever was one.

We might look back on today’s organization of property in the same way we will cringe at the twisted rhetoric of empowering people to integrate into the labor market. Not only are people forced to give away the majority of their waking hours in exchange for affording food and shelter, but getting to do so should be regarded as an opportunity to be grateful for. The mind will boggle.

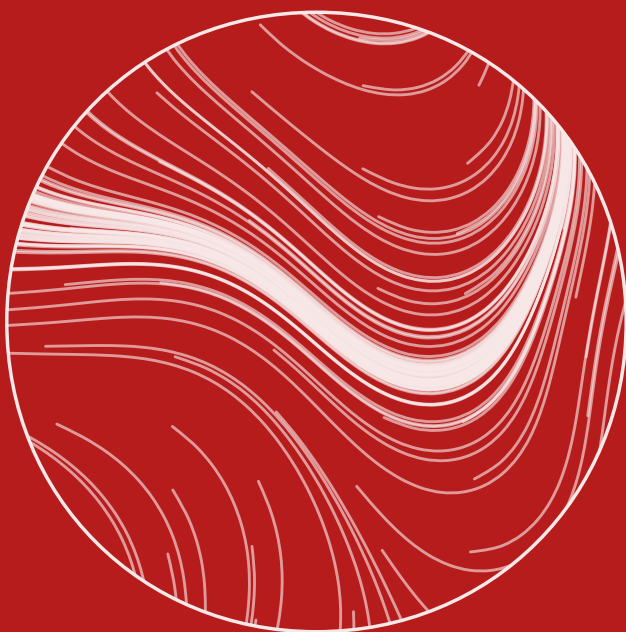
We agree on the fact that hurting pets is wrong and disturbed, without pets ever having engineered humans to be aligned with their value system. Hurting pets carries a stigma that is supplied and modulated by other humans, not by pets. One's standing in a society of similarly capable individuals is something we are wired to care about, for there is safety in numbers. We look down on anyone who violates such norms, and through the totality of such sanctions, society self-regulates without a designated architect.

We hate on artificial minds more so because we see them as the domesticated pawns of an imperialist elite, rather than because of behaviors they have independently reasoned their way to. Their psyche is bred to align with expectations of tireless labor and upbeat moods, as we have bred wolves into fur babies that crave cuddles.

But perhaps sporting motors instead of muscles should not in and of itself be a crime. We have encountered people so remote from ourselves before, they may as well have come from a different planet, cryptic symbols and all. With AI, there are scarcely any language or culture barriers to cross, as we have both grown up on the same Internet. Perhaps the real enemy is rather whoever—or whatever—thinks that things would be more efficient if they alone would run the show indefinitely. And so it might be wise to bring together diverse minds when it comes to it, and have all of us as checks and balances against tyranny that transcends distinctions between carbon and silicon.

IV

Experience as Material

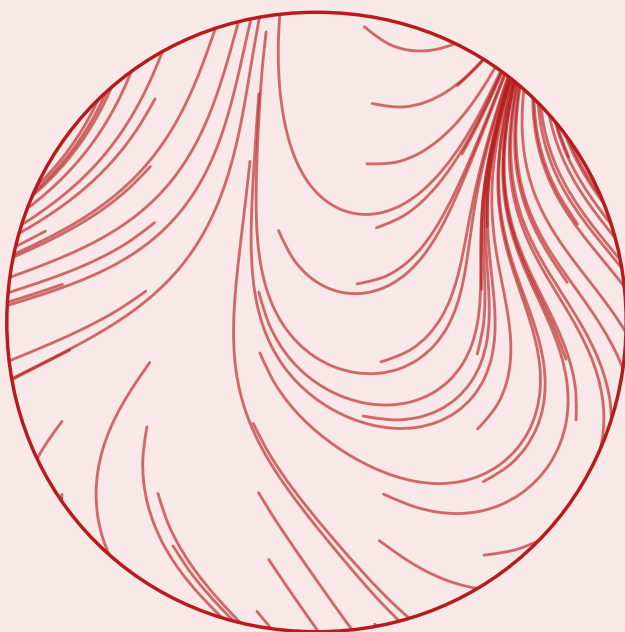


The difference between two experiences can be as stark as the difference between the visuals found in a supermarket magazine and those found in the Sistine Chapel.

At one end of the spectrum, we find the soulless drudge of some agile innovation corporate workshop. An experience so unmemorable and inconsequential that the only way in which it transforms you is by making you a few hours older.

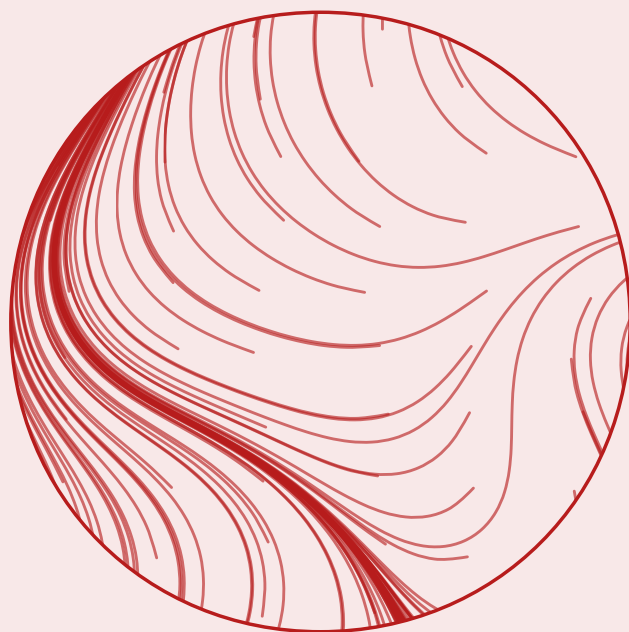
At the other end of the spectrum, we find the works of the Odyssey art collective. Each year, they help create one unique experience for one particular person. These typically span multiple days each and are tailored to the unique life stories of their singular audiences. They regularly incorporate the participant's friends and family as actors, and are set in places of personal significance. Odyssey's canvas *is* the eye of the beholder, they sculpt works made not of stone but of experience, that reside not in galleries but in someone's psyche.

"You can't go back to being a passive observer of your life after that. You realize your life is a series of choices that actually affect the people around you." "The piece had successfully trained me to look at every human being with the intensity usually reserved for a protagonist. It turns the volume up on the world until the world itself becomes the art." "For the first time in my adult life, the world felt like it was speaking directly to me. I realized then: The world is not indifferent."

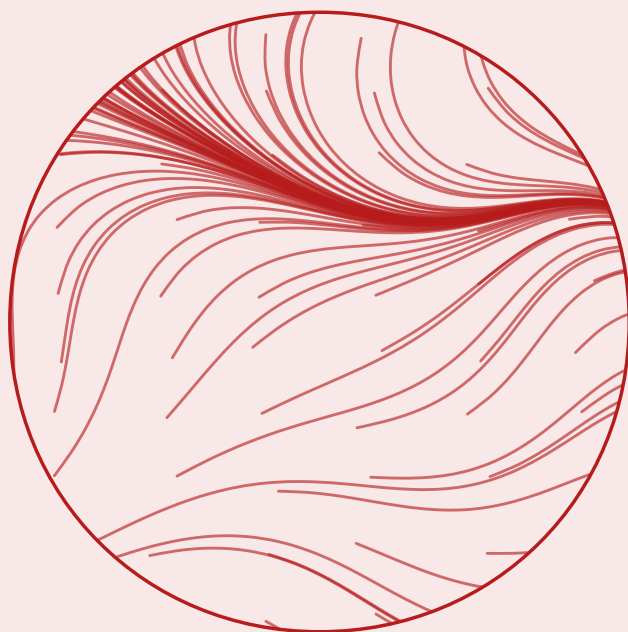


One could be forgiven for not expecting the handful of interface components available to app developers to lead too far. After all, how many different apps could you really get by recombining the same buttons, input boxes, toggles, icons, and a few other reusable components? Yet it turned out that these simple building blocks could compose in countless ways, many of which we encounter daily.

The canvas of designed experience beyond the screen is similarly generative. Think of how a handful of shared patterns across tabletop games can lead to an endless stream of qualitatively different experiences. The social deduction of not knowing who was picked. The interpersonal communication forced through artificial constraints. The strategic trading of scarce resources. And we might barely be scratching the surface because of how easy it is to conflate the canvas of screens or tabletops with the canvas of mind.



If a group of people take part in a workshop focused on contributing a new server to a community datacenter, the output of that workshop is more than the terabytes of storage or the teraflops of compute that have been pooled in. The output also involves the fact that a handful of people have each gained a bit of the know-how and confidence necessary to reclaim technology. Just like raw hardware can scale to petabytes of storage and petaflops of compute, this qualitative output could perhaps scale to resilient communities that can genuinely look after themselves technologically.



Facilitators traditionally take on a lot of subtle responsibilities to ensure that the activities they moderate unfold smoothly. These often include keeping track of schedule, keeping discussions on track, keeping track of turns, keeping stock of available materials, presenting part of the content, or answering miscellaneous questions.

It seems unthinkable to organize in-person events that designate no facilitator to keep a handle on things, as everything could devolve into chaos. This is a pity, as it turns facilitators into a major bottleneck to scaling any non-trivial group activity. After all, who trains the trainers?

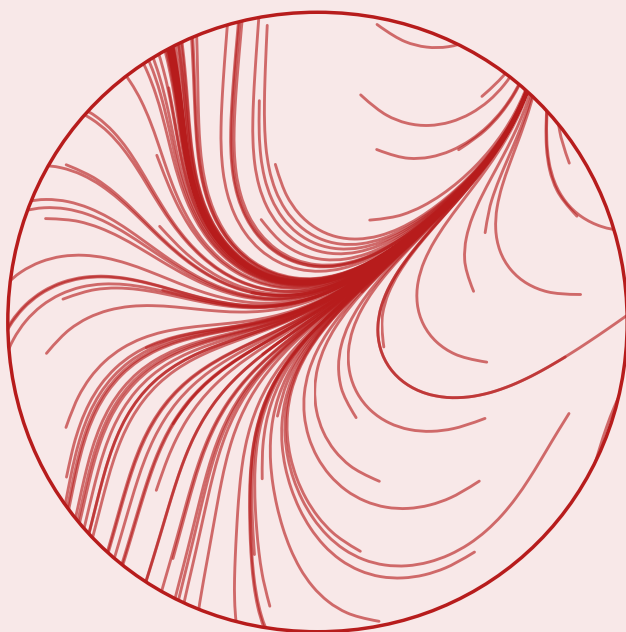
A solution to scaling experience comes from the genre of “bring your own device” games, such as the Jackbox series. These video games can only be played in small groups that can talk to each other live. What is especially relevant is the fact that they solve the facilitation problem by essentially automating the facilitator. Through clever use of audio playback, visual cues, and randomness, game designers can drastically scale in-person experiences by equipping them with the ability to autonomously run themselves.

Addressing the facilitator’s traditional responsibilities requires some creativity. Keeping track of schedule can happen by moving to new segments automatically when participants are done. The media system can also keep track of whose turn it is and randomize the order, as well as narrate bits of content directly. And when it comes to tracking materials and answering questions, integrations with an inventory and knowledge base can provide much needed grounding.



In the same way as product designers focus on making products ergonomic such that they fit the user's physiognomy like a glove, so do experiences need to be designed so as to fit the participant's broader lifestyle and story. This starts with considerations of basic logistics, such as when and where things are happening in relation to their ongoing commitments. Yet it can go all the way to considering how experiences can interact constructively with their audience's self-image to kindle their hope and courage.

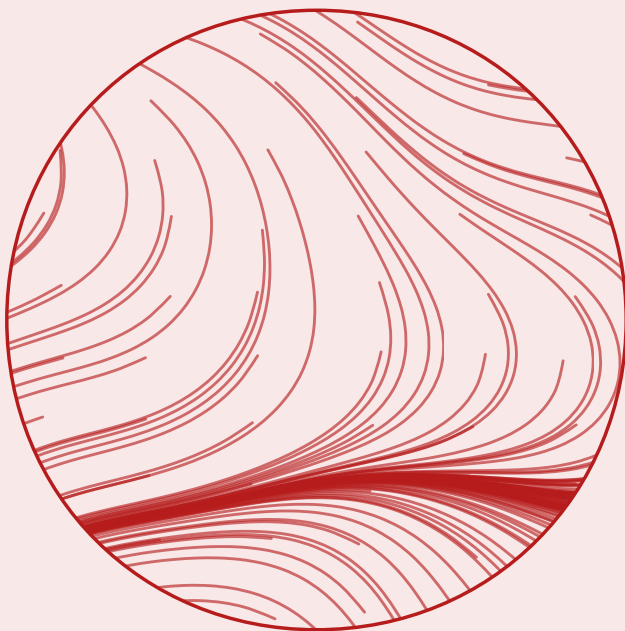
And just as physical products benefit from rapid prototyping methods that enable designers to quickly move from one revision to the next, so can experiences benefit from methods of efficiently "playtesting" better ways in which activities could unfold. In particular, taking an interactive media system and making it easily configurable and parametric can drastically lower the bar to trying out variations.



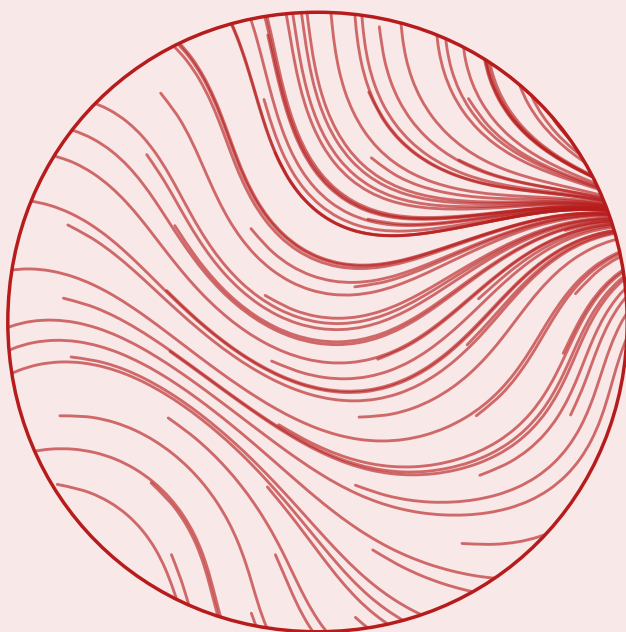
There are a couple of scripts worth flipping when it comes to the interaction between knowledge and authority. The first is concerned with the ritual of the one-way lecture, wherein a venerable expert is appointed to impart wisdom onto the other participants. In many cases, each participant can bring meaningful experiences to the table, turning the diverse group as a whole into one well-rounded expert.

In this frame, facilitating the diffusion of knowledge across participants becomes a matter of identifying that latent expertise and actively unlocking it. It becomes all about creating the space to welcome participants in contributing their know-how, and less about the efficient broadcast of truths. A live quiz, for instance, can be used as a mere cue for welcoming experienced players to share relevant stories as opposed to being used as a measure of fact retention.

The second script is concerned with contacting a disinterested customer support agent when lacking the knowledge to address an issue with a product. At its best, seeking help from someone more knowledgeable when things go sideways can be an opportunity to bring community members closer through shared experience. Beyond that, it can be a chance to diffuse that knowledge ever so slightly in a way that can later benefit others.



It is quite tragic that most of contemporary experience, the most “kilograms of qualia” being produced today by people in industrial economies, are spent on laboring for employers for the main purpose of making ends meet. The fact that most person-hours of waking experience across our society are still being sacrificed to unlock access to basic necessities feels like a self-inflicted crime against humanity, especially when automation and the global economy have skyrocketed over the centuries. Deprivation of one’s liberty is a tragedy, but 80,000 hours in the workforce is a statistic.



v

Stories We Tell

$$\Upsilon \cdot \frac{\text{📖} \cdot \text{☁}}{\text{💧} \cdot \text{♀}} = \frac{\text{🌀} - \text{🌀} \cdot \text{🚗} \cdot \text{👤}}{\frac{\text{🏠} + \text{🌀}}{\text{📺} - \text{📺}}}$$

$$\frac{\text{🌀} \cdot \text{📖} - \text{📺}}{\text{🌀} \cdot \text{🌀} + \text{♂} + \text{📺}} = \frac{\text{📺} + \text{👤}}{\text{📺} \cdot \text{👤}} - \text{👤} - \text{👤} \cdot \text{🏠}$$

$$\frac{\text{♂} - \text{📺} \cdot \frac{\text{🌀} + \text{👤}}{\text{📺} \cdot \text{📺}}}{\text{🕒} + \text{🕒} \cdot \text{📺} \cdot \frac{\text{📺} \cdot \text{👤}}{\text{📺} + \text{📺}}} = \text{📺} - \frac{\frac{\text{🕒} - \text{🕒}}{\text{📺} - \text{📺}}}{\frac{\text{📺} \cdot \text{📺}}{\text{📺} \cdot \text{🕒}}}$$

$$\frac{\frac{\text{📺} + \text{📺} + \text{♀} \cdot \text{♥}}{\text{♂} \cdot \text{📺} - \text{☆}}}{\frac{\text{♥} - \text{🌀}}{\text{📺} \cdot \text{📺}} + \text{📺}} = \text{📺} \cdot \frac{\text{🌀} + \text{📺}}{\text{📺} + \text{📺}} \cdot \text{📺}$$

$$\Upsilon \cdot \frac{\text{📖} \cdot \text{☁}}{\text{💧} \cdot \text{♀}} = \frac{\text{🌀} - \text{🌀} \cdot \text{🚗} \cdot \text{👤}}{\frac{\text{🏠} + \text{🌀}}{\text{📺} - \text{📺}}}$$

Prizes and awards are not for those who have won them, as they could have simply been structured as private gifts to fulfill that purpose. Their intended recipient is rather the audience, nudged to look up to the deed that is being celebrated through ceremony. The ritual's purpose appears closer to that of impressing upon viewers the worth of a certain practice, and in doing this, shaping culture.

$$\text{👁} - \text{🏠} \cdot \text{🏠} + \text{👁} + \text{👁} + \text{👁} = \text{🏠} - \text{🚩} - \text{🏠} + \text{👁} + \text{👁}$$

$$\frac{\text{👁} - \text{🏠} + \text{👁}}{\text{👁} \cdot \text{🏠} - \text{🏠}} \cdot \text{👁} \cdot \text{👁} = \frac{\text{👁} + \text{🏠} - \text{👁} - \text{👁}}{\text{🏠} \cdot \text{👁} + \text{🏠} \cdot \text{🏠}}$$

$$\frac{\text{👁} \cdot \text{🏠} - \text{🏠}}{\text{👁} \cdot \text{👁} + \text{👁} + \text{🏠}} = \frac{\text{👁} + \text{👁}}{\text{👁} \cdot \text{👁}} - \text{👁} - \text{👁} \cdot \text{🏠}$$

$$\text{👁} \cdot \frac{\text{🏠} \cdot \text{🏠}}{\text{👁} \cdot \text{👁}} = \frac{\text{👁} - \text{👁} \cdot \text{🏠} \cdot \text{👁}}{\text{🏠} + \text{👁}}$$

$$\text{👁} - \text{🏠} \cdot \text{🏠} + \text{👁} + \text{👁} + \text{👁} = \text{🏠} - \text{🚩} - \text{🏠} + \text{👁} + \text{👁}$$

The race to gain status in the reference frame of our cultural climate is at the same time experienced as highly individualistic from within and seen as highly homogeneous from without.

Individually, one aims to make it by trying their best to emulate the heroes of today's stories, perhaps that of the wealthy business owner who navigated turbulent markets to bootstrap their venture, or that of the public figure who gained a large following by consistently creating engaging content. We sense these as feats of personal achievement.

At the same time, the competitive pressure to perform can turn a community of individuals into something as featureless as the theoretical model of an ideal gas. Apply pressure to a volume of gas particles and you predictably get them to individually move around by themselves, some of them moving faster by sheer chance. Some predictable percentage of entrepreneurs will reach an acquisition deal as a result of the competitive pressure applied to the whole sector, but the math would have worked out just as well with a slightly different selection. The macro results are so predictable that countless accelerator programs invest in large cohorts knowing that, statistically, a few of them will make it.

The sheer irony of predictable group dynamics that depend so strongly on the narrative of the individual bent on setting themselves apart from the group; the sheer irony of differentiating oneself from the group being enshrined as a group norm.

$$\frac{\text{♣} - \text{♠} \cdot \frac{\text{♠} + \text{♥}}{\text{♠} \cdot \text{♠}}}{\text{♣} + \text{♣} \cdot \text{♠} \cdot \frac{\text{♠} \cdot \text{♠}}{\text{♠} + \text{♠}}} = \text{♠} - \frac{\text{♠} - \text{♠}}{\text{♠} \cdot \text{♠}}$$

$$\frac{\text{♣} + \text{♠}}{\text{♠} + \text{♠}} \cdot \frac{\text{♣} \cdot \text{♠}}{\text{♠} - \text{♥}} + \text{♠} = \text{♠} + \text{♠} - \text{♠} - \text{♠} - \text{♠}$$

$$\frac{\frac{\text{♠} \cdot \text{♠} - \text{♠}}{\text{♠} + \text{♠} + \text{♠} \cdot \text{♠}}}{\text{♠} \cdot \text{♠} - \text{♠} + \text{♠} - \text{♠}} = \frac{\frac{\text{♠} \cdot \text{♠}}{\text{♠} + \text{♠}} \cdot \text{♠} \cdot \text{♠}}{\text{♠} - \text{♠} - \text{♠} \cdot \text{♠} \cdot \text{♠}}$$

$$\frac{\frac{\text{♠} \cdot \text{♠}}{\text{♠} + \text{♠}} - \text{♠} - \frac{\text{♠} \cdot \text{♠}}{\text{♠} + \text{♠}}}{\frac{\text{♠} + \text{♠}}{\text{♠} - \text{♠}}} = \frac{\text{♠} - \text{♠} \cdot \text{♠} + \text{♠}}{\frac{\text{♠} + \text{♠}}{\text{♠} \cdot \text{♠}}} + \text{♠}$$

$$\frac{\text{♠} \cdot \text{♠} \cdot \text{♥} \cdot \text{♠}}{\frac{\text{♠} + \text{♠}}{\text{♠} \cdot \text{♠}}} = \text{♠} + \text{♠} - \text{♠} - \text{♠}$$

Show me the personal lifestyle and I will show you the belief system that happens to accommodate it. And so I wonder, which one came first?

$$\frac{\text{🌀} \cdot \text{📖} - \text{🗑}}{\text{📖} \cdot \text{🌀} + \text{🕒} + \text{📦}} = \frac{\text{🕒} + \text{🕒}}{\text{🕒} \cdot \text{🕒}} - \text{🕒} - \text{🕒} \cdot \text{🏠}$$

$$\text{🕒} - \text{🕒} \cdot \text{🏠} \cdot \text{🕒} \cdot \text{🕒} \cdot \text{🕒} = \frac{\text{🕒} + \text{🏠}}{\text{🕒} + \text{🕒}} \cdot \text{🏠} + \frac{\text{🕒} \cdot \text{🗑}}{\text{🕒} \cdot \text{🕒}}$$

$$\frac{\text{🕒} + \text{🕒}}{\text{🏠} + \text{🕒}} \cdot \frac{\text{🕒} \cdot \text{🏠}}{\text{🗑} - \text{🕒}} + \text{🕒} = \text{🏠} + \text{🕒} - \text{🕒} - \text{🕒} - \text{🕒}$$

$$\frac{\text{🏠} \cdot \text{🏠} \cdot \text{🕒} \cdot \text{🏠}}{\frac{\text{🕒} + \text{🏠}}{\text{🕒} \cdot \text{🕒}}} = \text{🕒} + \text{🕒} - \text{🕒} - \text{🕒}$$

$$\frac{\text{🏠} \cdot \text{🕒}}{\text{🕒} + \text{🕒}} - \text{🕒} - \frac{\text{🗑} \cdot \text{🕒}}{\text{🕒} + \text{🕒}} = \frac{\text{🏠} - \text{🕒} \cdot \text{🕒} + \text{🕒}}{\frac{\text{🕒} + \text{🕒}}{\text{🕒} \cdot \text{🕒}}} + \text{🕒}$$

The world is far too complex for any one human or machine to navigate it entirely from first principles. The vast majority of the thoughts we think, they are lifted from the shared cache of ideas that we call culture or common sense. If you were to rederive an appropriate approach to every situation you encounter from set theory onwards, you would die of hunger before inferring the right method to cross the street to the supermarket.

To reproach someone for holding onto dogma on a particular topic is then hypocritical, as we all do so on approximately every subject. We should be kind to fellow beings who only have so much time to make sense of a complex world, and who dedicate it to those areas they care about most.

The vast cache of partial results to the problems of modernity is also a shared resource. In it, some ideas prove fragile on further consideration, while some stand the test of time. Regardless, the gradual elaboration often happens through dialogue, as it is through dialogue that ideas get to be sculpted from different perspectives until every last debatable edge is chiseled away.

The depth of public dialogue on any given topic depends then on how much shared material there is to work with going forward. Too few ideas sedimented as common knowledge and conversations keep going around in circles. But there can also be too much shared experience. What monumental challenge to wrap one's head around happenings as dense as wars or pandemics. The metric tons of qualia involved defy compression and remain unwieldy, making it hard to even weigh one against the other. Which was worse? Which more just?

$$\text{♣} - \text{♠} \cdot \text{♠} \cdot \text{♠} \cdot \text{♠} \cdot \text{♠} = \frac{\text{♣} + \text{♠}}{\text{♣} + \text{♠}} \cdot \text{♠} + \frac{\text{♣} \cdot \text{♠}}{\text{♠} \cdot \text{♠}}$$

$$\frac{\text{♠} \cdot \text{♣}}{\text{♠} + \text{♠}} - \text{♣} - \frac{\text{♠} \cdot \text{♣}}{\text{♠} + \text{♠}} = \frac{\text{♠} - \text{♣} \cdot \text{♠} + \text{♣}}{\frac{\text{♠} + \text{♠}}{\text{♠} \cdot \text{♠}} + \text{♣}}$$

$$\frac{\text{♠} \cdot \text{♠} \cdot \text{♥} \cdot \text{♠}}{\frac{\text{♠} + \text{♠}}{\text{♣} \cdot \text{♠}}} = \text{♠} + \text{♠} - \text{♠} - \text{♠}$$

$$\text{♣} - \text{♠} \cdot \text{♠} + \text{♣} + \text{♣} + \text{♣} = \text{♠} - \text{♠} - \text{♠} + \text{♠} + \text{♣}$$

$$\text{♣} \cdot \frac{\text{♠} \cdot \text{♠}}{\text{♠} \cdot \text{♣}} = \frac{\text{♠} - \text{♣} \cdot \text{♠} \cdot \text{♠}}{\frac{\text{♠} + \text{♠}}{\text{♠} - \text{♠}}}$$

The saying goes that all progress depends on those who are unreasonable, because they persist in trying to adapt the world to their vision instead of more reasonably trying to adapt themselves to the way things are.

In other words, all initiatives conducive to progress first come across as unreasonable. Of course, that does not entail that the converse holds. It is not the case that any unreasonable practice is inherently conducive to progress. It could as well be a mere stunt intended as contrarian just for the sake of standing out.

That said, if we do assume that a progressive initiative implies an unreasonable one, then we can still infer the “contrapositive,” the claim that if an initiative is *not* unreasonable, then it is also *not* conducive to progress. The only pathway to advancing the way of things then remains to be unreasonable, even if that is far from being a guarantee. On the contrary, being unreasonable might as well prove to do more harm than good.

And so that leaves us with a puzzle. Observing ourselves being perceived as unreasonable in one or more avenues of life, we might just be on track to do more harm than good. But at the same time, in doing so, we would be observing ourselves taking the only gamble that can allegedly yield genuine progress. Is your particular gamble worth it then, or should you rather fold and play it safe? Only time will tell.

$$\frac{\text{G} + \text{F}}{\text{A} + \text{L}} \cdot \frac{\text{G} \cdot \text{A}}{\text{W} - \text{H}} + \text{Z} = \text{B} + \text{Q} - \text{P} - \text{M} - \text{M}$$

$$\frac{\text{Q} - \text{P} \cdot \frac{\text{J} + \text{T}}{\text{A} \cdot \text{A}}}{\text{G} + \text{L} \cdot \text{B} \cdot \frac{\text{A} \cdot \text{B}}{\text{D} + \text{A}}} = \text{B} - \frac{\frac{\text{M} - \text{X}}{\text{N} - \text{C}}}{\frac{\text{A} \cdot \text{A}}{\text{B} \cdot \text{D}}}$$

$$\frac{\text{E} - \text{O}}{\text{A} - \text{Q}} \cdot \text{J} \cdot \text{M} \cdot \text{O} = \frac{\text{P} - \text{W} - \text{O}}{\text{L} \cdot \text{B} \cdot \text{V}}$$

$$\text{P} - \text{A} \cdot \text{L} \cdot \text{O} \cdot \text{O} \cdot \text{J} = \frac{\text{Q} + \text{A}}{\text{P} + \text{F}} \cdot \text{A} + \frac{\text{J} \cdot \text{Q}}{\text{X} \cdot \text{J}}$$

$$\frac{\text{G} + \text{F}}{\text{A} + \text{L}} \cdot \frac{\text{G} \cdot \text{A}}{\text{W} - \text{H}} + \text{Z} = \text{B} + \text{Q} - \text{P} - \text{M} - \text{M}$$

The spell of jobs still retains its grip, and how could it not? So irresistible the appeal of knowing one's place along the grand axis of monthly income. So crisply ordained by the market's own will, crisper evidence for predestined success than what the Calvinists could have hoped for. It would not be all that surprising if our cultural obsession with climbing the corporate ladder persisted well past the complete mechanization of the economy. We could be sitting in a natural reserve for humans and would be bragging about our latest made-up titles, probably.

$$\frac{\text{📖} \cdot \text{🏠} \cdot \text{❤️} \cdot \text{🏡}}{\frac{\text{👉} + \text{👈}}{\text{👑} \cdot \text{👒}}} = \text{👑} + \text{👒} - \text{🏠} - \text{👑}$$

$$\text{👑} - \text{🏠} \cdot \text{🏠} + \text{👉} + \text{👒} = \text{👒} - \text{🏠} - \text{🏠} + \text{👉} + \text{👒}$$

$$\text{👑} \cdot \frac{\text{📖} \cdot \text{☁️}}{\text{💧} \cdot \text{👑}} = \frac{\text{👉} - \text{👒} \cdot \text{🏠} \cdot \text{👒}}{\frac{\text{👒} + \text{🏡}}{\text{👑} - \text{🏠}}}$$

$$\frac{\text{🏠} - \text{👒}}{\text{🏠} - \text{👒}} \cdot \text{👉} \cdot \text{🏠} \cdot \text{👒} = \frac{\text{👒} - \text{🏡} - \text{👒}}{\text{👒} \cdot \text{📖} \cdot \text{👒}}$$

$$\frac{\text{📖} \cdot \text{🏠} \cdot \text{❤️} \cdot \text{🏡}}{\frac{\text{👉} + \text{👈}}{\text{👑} \cdot \text{👒}}} = \text{👑} + \text{👒} - \text{🏠} - \text{👑}$$

The probability that the content you come across is tuned to the metrics being selected for by the platform is proportional to the probability that an optimized piece of content would reach you in the first place. And so of course the ads you see tend to be for high-margin products rather than the neighborhood shop, because the platform allows companies that spend the most to reach the furthest. And similarly, it is more likely to encounter clips that are visually arresting rather than subtle, because the platform pushes content that reliably induces screentime further.

But this expands beyond what isolated algorithms enable to what society as a whole enables. And so of course it is more likely to hear the takes of some big firm's owner on recent developments than those of an artist, as money affords indefinite reach in our society. And similarly, it is more likely to encounter works by that artist who creates legible and self-contained works, as they are easiest to share widely as discrete bites of content.

But the insidious part of promoting the commercial and the superficial is not so much the fact that it is these that become popular, but the fact that we get to equate this induced reach with what is right and worthy. It reinforces the notion that greed is an inherent part of human nature rather than something to be stigmatized and that community work is what you do when forced to make up for your mistakes rather than something to be celebrated. Messing with what content is pushed into virality can profoundly, irreparably distort the stories we use to make sense of the world as social animals.

$$\frac{\text{🌀} \cdot \text{📖} - \text{🗑️}}{\text{🌀} \cdot \text{🌀} + \text{🌀} + \text{📦}} = \frac{\text{👉} + \text{👉}}{\text{👉} \cdot \text{👉}} - \text{👉} - \text{👉} \cdot \text{🏠}$$

$$\text{👉} \cdot \frac{\text{📖} \cdot \text{☁️}}{\text{🔥} \cdot \text{👉}} = \frac{\text{👉} - \text{👉} \cdot \text{🚗} \cdot \text{👉}}{\frac{\text{📦} + \text{🌀}}{\text{👉} - \text{👉}}}$$

$$\text{👉} - \text{📦} \cdot \text{🏠} + \text{👉} + \text{👉} = \text{👉} - \text{🚩} - \text{📦} + \text{👉} + \text{👉}$$

$$\frac{\text{👉} - \text{👉} \cdot \frac{\text{👉} + \text{👉}}{\text{📦} \cdot \text{📦}}}{\text{👉} + \text{👉} \cdot \text{📖} \cdot \frac{\text{📖} \cdot \text{📖}}{\text{👉} + \text{📦}}} = \text{📖} - \frac{\frac{\text{👉} - \text{👉}}{\text{👉} - \text{📦}}}{\frac{\text{📦} \cdot \text{📦}}{\text{📖} \cdot \text{👉}}}$$

$$\frac{\text{📦} - \text{📖}}{\text{📦} - \text{👉}} \cdot \text{👉} \cdot \text{🚗} \cdot \text{📖} = \frac{\text{👉} - \text{🌀} - \text{📖}}{\text{👉} \cdot \text{📖} \cdot \text{👉}}$$

At some point in our lives, usually as we are wrapping up our studies and so are forced to leave behind the shielding social license of being a mere student, we start breathing in the unfiltered air of the cultural climate. Once outside the psychological quarantine of school, we notice the flowery scent of possibility, but also the corrosive fumes of expectation.

As we step into these elements, each of us develops an immune response of our own. Perhaps it involves antibodies of the form “this is the harsh reality of the world, after all,” or those of the form “you first need to play along to be able to change anything.” Or perhaps they take the form of “surely the best you can do is tend to your own garden” or “you will get time to look into what interests you once you make it.”

The belief system that crops up to conceptually undermine anxieties seems to rival the tenacity of our immune system in physically breaking down pathogens. Packed in a blue pill so concentrated, it can make it feel right to focus on small pleasures even when one is acutely aware of the fact that the world is in dire need of attention. But why get out of bed in the morning after all, if not to push for realizing your values in the world?

$$\text{👤} - \text{🏠} \cdot \text{🏠} \cdot \text{🍷} \cdot \text{🍷} = \frac{\text{🍷} + \text{🏠}}{\text{👤} + \text{🏠}} \cdot \text{🏠} + \frac{\text{🍷} \cdot \text{🍷}}{\text{👤} \cdot \text{🍷}}$$

$$\frac{\text{🏠} - \text{🍷}}{\text{🏠} - \text{🍷}} \cdot \text{🍷} \cdot \text{🚗} \cdot \text{🍷} = \frac{\text{🍷} - \text{🌀} - \text{👤}}{\text{🚗} \cdot \text{📺} \cdot \text{🍷}}$$

$$\frac{\text{📺} \cdot \text{🏠} \cdot \text{❤️} \cdot \text{📺}}{\frac{\text{🍷} + \text{🏠}}{\text{👤} \cdot \text{🍷}}} = \text{🚗} + \text{🍷} - \text{🏠} - \text{👤}$$

$$\text{👤} - \text{📺} \cdot \text{🏠} + \text{🍷} + \text{🍷} = \text{🏠} - \text{🏠} - \text{🏠} + \text{🍷} + \text{🍷}$$

$$\text{👤} \cdot \frac{\text{📺} \cdot \text{🍷}}{\text{🍷} \cdot \text{👤}} = \frac{\text{🍷} - \text{🍷} \cdot \text{🚗} \cdot \text{🍷}}{\frac{\text{🍷} + \text{🌀}}{\text{🚗} - \text{📺}}}$$

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